

Sidon sets in the squares, repeated distances, and the Elekes–Rónyai problem

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Two classical facts about sums of two squares

Fact 1 (Landau–Ramanujan)

$$\#\{m \leq x : m = a^2 + b^2\} = \Theta\left(\frac{x}{\sqrt{\log x}}\right).$$

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Fact 2 (Ramanujan/Euler/Gauss/Jacobi...)

$$\max_{m \leq x} r_2(m) = \exp\left((\log 2 + o(1)) \frac{\log x}{\log \log x}\right).$$

Here $r_2(m) = \#\{(a, b) \in \mathbb{Z}^2 \text{ so that } m = a^2 + b^2\}$.

- If $p \equiv 1 \pmod{4}$, then $r_2(p) = 8$.
- If $m = p_1^{e_1} \cdots p_t^{e_t} q_1^{f_1} \cdots q_u^{f_u}$, where $p_i \equiv 1 \pmod{4}$ and $q_j \equiv 3 \pmod{4}$, then

$$r_2(m) = 4(1 + e_1) \cdots (1 + e_t).$$

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$$r_2(m) = 4(1 + e_1) \cdots (1 + e_t).$$

Pick $p_i \in [\log x, 2 \log x]$ and $t \asymp \log x / \log \log x$. Then, $p_1 \cdots p_t \leq x$ and

$$r_2(p_1 \cdots p_t) = 4 \cdot 2^t \geq \exp\left(c \frac{\log x}{\log \log x}\right).$$

Erdős observation # 1: distinct distances

Every squared distance determined by $[\sqrt{n}] \times [\sqrt{n}]$ has the form

$$a^2 + b^2 \leq 2n.$$

Conversely, every $m \leq n/4$ that is a sum of two squares occurs as a squared distance.

Observation

$$\#\{\text{distinct distances determined by } [\sqrt{n}]^2\} = \Theta\left(\frac{n}{\sqrt{\log n}}\right).$$

Let

$$D(n) := \min_{|P|=n} \#\{|p - p'| : p, p' \in P, p \neq p'\}.$$

The distinct distances conjecture

$$D(n) \gtrsim \frac{n}{\sqrt{\log n}}.$$

- Guth-Katz ('10): $D(n) \gtrsim n/\log n$.

Erdős observation # 2: one popular distance

Observation

Some distance in $[\sqrt{n}] \times [\sqrt{n}]$ is repeated $\gtrsim n \exp\left(c \frac{\log n}{\log \log n}\right)$ times.

Let

$$U(n) := \max_{|P|=n} \#\{\{p, p'\} \in \binom{P}{2} : |p - p'| = 1\}.$$

- Scaling gives $U(n) \gtrsim n \exp\left(c \frac{\log n}{\log \log n}\right)$.

The unit distance conjecture

$$U(n) = n^{1+o(1)}.$$

- Best known upper bound: $n^{4/3}$ (Spencer-Szemerédi-Trotter, '84)

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$$U(n) = n^{1+o(1)}.$$

- Best known upper bound: $n^{4/3}$ (Spencer-Szemerédi-Trotter, '84)
- Disproved by OpenAI on May 20th, 2026:

There exists a set of n points where a distance repeats $\geq n^{1.01}$ times.

Other Erdős problems in the square grid

Problem 1 (Erdős–Guy, '70)

How large can $A \subset [\sqrt{n}]^2$ be if all distances determined by A are distinct?

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Since the grid determines only $n/\sqrt{\log n}$ possible distances, we must have

$$\binom{|A|}{2} \lesssim \frac{n}{\sqrt{\log n}} \implies |A| \lesssim \frac{\sqrt{n}}{(\log n)^{1/4}}.$$

- Lower bound: $n^{1/3}$

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Problem 2 (Erdős, '80)

How large can $A \subset [\sqrt{n}] \times [\sqrt{n}]$ be if it contains no isosceles triangle?

The number of distinct distances determined by A is at least $|A| - 1$, so

$$|A| \lesssim \frac{n}{\sqrt{\log n}}.$$

- Lower bound: $\sqrt{n}$

Can one improve on any of these?

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Each vertical grid line intersects A in a 3AP-free subset, so

$$|A| \leq \sqrt{n} \cdot r_3(\sqrt{n}) \leq \frac{n}{\exp(c(\log n)^{1/5})}.$$

- Lower bound: $\sqrt{n}$

Can one improve on any of these?

A robust Ramanujan estimate

For a finite $A \subset \mathbb{R}^2$, let

$$\mu(A) := \max_{\lambda > 0} \#\{(a, b) \in A^2 : a \neq b, |a - b| = \lambda\}.$$

Theorem (Croot–Mao–P.–Sheffer–Yip, '26)

Every $A \subseteq [\sqrt{n}]^2$ with $|A| \geq 2$ satisfies

$$\mu(A) \gtrsim \frac{|A|^2}{n} \exp\left(c \frac{\log n}{\log \log n}\right).$$

When $A = [\sqrt{n}]^2$, this is the classical Ramanujan estimate.

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- If distances do not repeat in A , then $\mu(A) = 2$, so $|A| \lesssim \sqrt{n} / \exp\left(c \frac{\log n}{\log \log n}\right)$.

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When $A = [\sqrt{n}]^2$, this is the classical Ramanujan estimate.

- If distances do not repeat in A , then $\mu(A) = 2$, so $|A| \lesssim \sqrt{n} / \exp\left(c \frac{\log n}{\log \log n}\right)$.
- If A has no isosceles triangle, then $\mu(A) \leq |A|$, so $|A| \lesssim n / \exp\left(c \frac{\log n}{\log \log n}\right)$.

Vertical amplification

Theorem (Lee–P.–Zhu, 2026)

There exists an absolute constant $\delta > 0$ such that the following holds: For every n there is a set of n points $P \subset \mathbb{R}^2$ such that

$$\mu(A) \gtrsim \frac{|A|^2}{n^{1-\delta}}$$

for every $A \subseteq P$ with $|A| \geq 2$.

For this one set P :

- ❶ some distance occurs at least $n^{1+\delta}$ times;
- ❷ every $|A| \gtrsim n^{1-\delta}$ contains an isosceles triangle;
- ❸ every $|A| \gtrsim n^{1/2-\delta}$ repeats a distance.

The existence of a set with Property 2 confirms a conjecture of Erdős from 1980.

The existence of a set with Property 3 answers a question of Conlon, Fox, Gasarch, Harris, Ulrich and Zbarsky from 2015 in the negative.

Sidon sets inside the perfect squares

Question (Erdős, 1980s)

What is the largest Sidon set

$$A \subseteq \{1^2, 2^2, \dots, N^2\}?$$

Recall: A set A is *Sidon* if

$$a + b = c + d \quad (a, b, c, d \in A) \implies \{a, b\} = \{c, d\}.$$

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Since $A + A$ consists of sums of two squares in $[2N^2]$,

$$|A|^2 \lesssim |A + A| \lesssim \frac{N^2}{\sqrt{\log N}} \implies |A| \lesssim \frac{N}{(\log N)^{1/4}}.$$

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- Hanson ('20): $|A| \lesssim N/(\log N)^{1/2}$.
- Alon-Erdős, Thiele: $|A| \gtrsim N^{2/3}$.

A new upper bound

Theorem (Croot–Mao–P.–Sheffer–Yip, 2026)

Every Sidon set $A \subseteq \{1^2, 2^2, \dots, N^2\}$ satisfies

$$|A| \leq N \exp \left(-c \frac{\log N}{\log \log N} \right).$$

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Key point: squares occupy only $\frac{p+1}{2}$ residue classes modulo every odd prime p .

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Helfgott-Venkatesh: say that $A \subseteq [M]$ is α -ill-distributed if

$$|\{a \bmod p : a \in A\}| \leq \alpha p \text{ for every prime } p.$$

Theorem (Croot–Mao–P.–Sheffer–Yip, 2026)

If $A \subseteq [M]$ is Sidon and α -ill-distributed, then

$$|A| \leq \sqrt{M} \exp \left(-c \log(1/\alpha) \frac{\log M}{\log \log M} \right).$$

Proof: count congruence collisions

Choose primes $p_1, \dots, p_t \in [\log N, 2 \log N]$ with $t \asymp \frac{\log N}{\log \log N}$. Again, we have that $d = p_1 \cdots p_t \leq N$.

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Congruence energy

$$\Psi(A) := \#\{(a, b) \in A^2 : a \equiv b \pmod{d}\}.$$

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If A is Sidon, each nonzero difference $a - b$ occurs at most once. Therefore

$$\Psi(A) \leq |A| + \frac{2N^2}{d} \lesssim \frac{N^2}{d}.$$

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To lower bound $\Psi(A)$, write $A = \bigsqcup_{\lambda \bmod d} A(\lambda)$, where

$$A(\lambda) = \{x \in A : x \equiv \lambda \pmod{d}\}.$$

Then,

$$\Psi(A) = \sum_{\lambda} |A(\lambda)|^2$$

Proof: squares occupy few residue classes

By Cauchy–Schwarz,

$$\Psi(A) = \sum_{\lambda} |A(\lambda)|^2 \geq \frac{1}{r} \left(\sum_{\lambda} |A(\lambda)| \right)^2 = \frac{|A|^2}{r},$$

where r is the number of occupied residue classes modulo d .

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Arithmetic input

Squares occupy $(p_i + 1)/2$ classes modulo p_i , so the Chinese remainder theorem gives

$$r \leq \prod_{i=1}^t \frac{p_i + 1}{2} = \frac{d}{2^t} \prod_{i=1}^t \left(1 + \frac{1}{p_i} \right) \leq \frac{d}{2^t} (1 + o(1)).$$

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Combining the two estimates for $\Psi(A)$,

$$\frac{|A|^2 2^t}{d} \lesssim \frac{N^2}{d}, \quad |A| \lesssim \frac{N}{2^{t/2}} = \frac{N}{\exp\left(c \frac{\log N}{\log \log N}\right)}.$$

B_3 -sets in the cubes and B_4 -sets in the fourth powers

Recall: A set A is a B_k -set if

$$a_1 + \dots = a_k = b_1 + \dots + b_k \implies \{a_1, \dots, a_k\} = \{b_1, \dots, b_k\}$$

Theorem (Croot–Mao–P.–Sheffer–Yip, 2026)

Every B_3 -set $A \subseteq \{1, 2^3, \dots, N^3\}$ satisfies

$$|A| \lesssim N \exp\left(-c \frac{\sqrt{\log N}}{\log \log N}\right).$$

- This is the first $o(N)$ bound for this problem.
- Unlike sums of two squares, sums of three cubes are conjectured to occupy a positive proportion of the integers, so can't get $o(N)$ from $A + A + A \subset [3N^3]$.

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- Unlike sums of two squares, sums of three cubes are conjectured to occupy a positive proportion of the integers, so can't get $o(N)$ from $A + A + A \subset [3N^3]$.
- The proof uses an entropic version of the previous sieve, with local arithmetic input from the Fermat cubic over finite fields.
- CMPSY ('26): Every B_4 -set $A \subseteq \{1, 2^4, \dots, N^4\}$ satisfies

$$|A| \lesssim \frac{N}{(\log \log N)^c}.$$

The Elekes–Rónyai problem

Let $f \in \mathbb{R}[x, y]$ have fixed degree and let $|A| = |B| = n$. How small can

$$|f(A, B)| = |\{f(a, b) : a \in A, b \in B\}|$$

be?

- Small image sets are possible for

$$f(x, y) = g(h(x) + k(y)) \quad \text{or} \quad f(x, y) = g(h(x)k(y)).$$

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Conjecture (Elekes, '90s)

If f is neither additive, nor multiplicative, then

$$|f(A, B)| \geq n^{2-o(1)}$$

holds for all sets $A, B \subset \mathbb{R}$ with $|A| = |B| = n$.

- Elekes–Rónyai ('00): $\omega(n)$
- Raz–Sharir–Solymosi ('16): $n^{4/3}$
- Solymosi–Zahl ('22): $n^{3/2}$

A counterexample

Theorem (P., 2026)

There is an absolute $c > 0$ and arbitrarily large sets $A \subset \mathbb{R}$ with

$$|\{x + y + (x - y)^2 : x, y \in A\}| \leq |A|^{2-c}.$$

- This gives a counterexample for the Elekes-Rónyai problem.
- Moreover, for every $\varepsilon > 0$ the sets may be chosen with

$$|A + A| \leq |A|^{1+\varepsilon}, \quad |f(A, A)| \leq |A|^{2-c(\varepsilon)}.$$

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Main idea: if $A = \{1/d, 2/d, \dots, N/d\}$ for an appropriate squarefree number d ,

$$|\{x + y + (x - y)^2 : x, y \in A\}| \leq \frac{N^2}{\exp\left(c \frac{\log N}{\log \log N}\right)}.$$

Horizontal amplification for Elekes-Rónyai

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Equivalently, if we let $f_d(x, y) = d(x + y) + (x - y)^2$, we want to say that

$$f_d([N], [N]) \leq N^2 \exp\left(-c \frac{\log N}{\log \log N}\right).$$

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- Let $p_1, \dots, p_t$ be primes in $[\log N, 2 \log N]$ with $t \asymp \frac{\log N}{\log \log N}$, and let

$$d = p_1 \cdots p_t \leq N.$$

- For each $p \mid d$, $f_d(x, y) \equiv (x - y)^2 \pmod{p}$.
- $f_d([N], [N])$ occupies only about $d/2^t$ residue classes modulo d .
- For $1 \leq x, y \leq N$, we have that

$$2d \leq d(x + y) + (x - y)^2 \leq 2N^2 + (N - 1)^2.$$

It follows that

$$|F_d([N], [N])| \lesssim \frac{N^2}{2^t} = N^2 \exp\left(-c \frac{\log N}{\log \log N}\right).$$

Thank you